PROMOFINDER AI — EXECUTIVE SOLUTION DOCUMENT

PROBLEM 1: AUTOMATED PROMO DISCOVERY & GROUNDED RANKING FOR MICRODRAMAS

1. Problem Framing & Leverage

Promo discovery is the highest-leverage growth engine in microdrama streaming: a compelling 30-second vertical cut on Instagram, TikTok, or YouTube cold-converts passive scrollers into paying subscribers. Manually reviewing 50+ episodes weekly per show creates severe editorial bottlenecks. **PromoFinder AI** replaces subjective manual hunting with automated screenplay parsing, heuristic candidate windowing, LLM narrative evaluation, and deterministic Python constraint verification.

2. Operational Assumptions

Assumption	Screenplay Verification & Engineering Rationale
Scene-Relative Timestamps	Verified across all 50 episodes: every scene resets to 00:00.00. Multi-scene cuts track independent segment clocks.
Windowing Dynamics	Full scenes average 1–3 min; pipeline extracts 20s–60s cuts (ideal: 25s–45s) anchored on early hooks or closing climaxes.
SFX as Supporting Signals	Labels like CLIFFHANGER SFX are treated as audio hints, not verdicts—unresolved dialogue tension takes priority.
Hallucination Safety	Strict substring matching runs on quoted lines ("...") against script source, ensuring zero invented dialogue.

3. System Architecture & End-to-End Pipeline

Workflow: parser.py (scene & timestamp extraction) → analyzer.py (windowing: full-scene, first-45s, climax) → llm_client.py (Groq LLaMA 3.3 / OpenAI / Gemini / Mock) → validator.py (deterministic duration, timestamp existence, graph dedup) → scorer.py (5D continuous scoring) → reporter.py.
Deployment: Vercel Web Dashboard (React/Tailwind) + Azure Basic B1 FastAPI Backend + Vercel Edge Reverse Proxy (/api/*) + Supabase PostgreSQL Database. Screenplay source viewer streams all 50 full scripts instantly via CDN episodes.json.

4. Explainable 5-Dimension Grounded Scoring Rubric

Dimension	Weight	Engine	Screenplay Grounding & Evaluation Criteria
Hook Strength	25%	AI + Grounded	First 3s trigger: dialogue onset (≤1.5s), emotional delivery (Cold, Defiant, Whispering), shocking reveal.
Cliffhanger Strength	30%	AI + Grounded	Unresolved tension at cut point: open questions, abrupt breaks, gasps, and supporting cliffhanger SFX stings.
Standalone Clarity	20%	AI + Grounded	Self-contained narrative: cold viewer understands conflict without prior lore or 50 episodes of backstory.
Emotional Impact	15%	AI + Grounded	Visceral stakes: density of drama keywords (betrayal, marriage, scars, murder, duke, blade, blood).
Duration Quality	10%	Deterministic	Continuous smooth decay curve centered at 35.0s sweet spot: 10.0 - 3.2×(|d - 35|/25)^1.4. Bounds: [20s, 60s].

PROMOFINDER AI — EVALUATION, TELEMETRY & AUDIT

QUALITY TELEMETRY, FAILURE AUDIT, ENGINEERING BOUNDARIES & AI TOOL REFLECTION

5. AI vs. Rule-Based Allocation (~65% Deterministic / ~35% LLM Reasoning)

Task Component	Allocation	Engineering Rationale
Timestamp & Duration Bounds	100% Deterministic	LLMs struggle with arithmetic; strict Python checks guarantee zero constraint violations.
Graph Deduplication (≥60% IoU)	100% Deterministic	Greedy maximal independent set algorithm eliminates redundant overlapping cuts algorithmically.
Dialogue Quote Verification	100% Deterministic	Regex substring validation checks all quoted lines exist verbatim in source screenplay.
Narrative Quality Scoring	LLM Reasoning	Evaluates psychological curiosity, character emotional arousal, and unresolved narrative pull.

6. Telemetry & Deduplication Funnel (Full 50-Episode Pack Execution)

- **Total Evaluated:** 218 raw candidate windows across all 50 episodes (100% episode coverage).
- **Validation Pass Rate: 56.0%** (122 candidates admitted to ranking pool).
- **Redundant Prune Rate: 44.0%** (96 redundant overlapping cuts pruned via graph deduplication).
- **Accepted Pool Constraint Compliance: 100.0%** (122 / 122 with 0 timestamp errors, 0 hallucinations, 0 duration violations).
- **Deduplication Graph Math:** 133 pairwise overlap collisions (≥60% IoU) pruned down to exactly 96 eliminated vertices.
- **Ranked Output:** Top 20 distinct, non-overlapping promo moments (Scores range: 7.591 to 8.534, mean: 7.914).

7. Documented System Limitations & Failure Modes

Limitation	System Impact	Engineering Mitigation
Short Scenes (<20s)	Ep 39 & Ep 44 produce 0 promo windows.	Intended: clips under 20s lack narrative weight for social ad retention.
Audio Delivery Blindness	Text scripts omit actor pacing & audio tone.	Emotional stage directions (e.g. (Coldly), (Whispering)) are parsed & scored.
Multi-Character Cross-Talk	Scenes with 4+ speakers confuse cold traffic.	Standalone clarity metric penalizes scenes exceeding 3 active speakers.

8. AI Tools Used & Documented AI Failure Case

AI Tools Utilized: Google Gemini API, OpenAI GPT-4o-mini, Groq LLaMA 3.3 70B, and Antigravity Assistant for architecture review.
Documented AI Failure During Development: Early in development, the AI-generated hallucination validator was prompted to check whether the entire generated candidate summary was a substring of the source script. This caused an catastrophic false-positive failure, rejecting **100% of valid candidates (218/218)** because AI rationales are descriptive paraphrases (e.g., *"Alexander confronts the matriarch"*) which do not appear literally in scripts. **Fix:** We re-engineered the validator to isolate verbatim dialogue quotes ("...") and verify only those strings against the script text, restoring 100% accuracy while preventing actual hallucinations.

9. Scalability to Future Shows

Zero code modifications required. Processing a new microdrama simply requires placing raw .txt files into data/scripts/<show_name> and executing python src/pipeline.py. The parser auto-detects scene headings, bracket timestamps, and ALL-CAPS script variants. For production editorial teams, the live Web Ingestion tab analyzes pasted screenplays or public URLs on-demand.